



Textovid

AI Comic Studio

Project Profile

A multimodal AI application that generates unique, original comic books from text prompts or category selections, combining LLM storytelling with SDXL image generation on AMD Radeon GPUs via ROCm 7.2.

Platform: AMD Radeon RX 7900 XTX (24 GB VRAM)

Software: PyTorch 2.11 + ROCm 7.2 + Gradio 6.20

Models: Qwen 35B (LLM) + SDXL + SDXL Refiner + LTX Video

ROCm 7.2

PyTorch

SDXL

Float16

xformers

Table of Contents

1. Project Overview	2
2. Target Users and Use Cases	2
2.1 Primary Audience	2
2.2 Creative Scenarios	2
2.3 Practical Value	3
3. System Architecture	3
3.1 Pipeline Overview	3
3.2 Module Breakdown	3
4. Model and Algorithm Introduction	4
4.1 Text Generation: Qwen 35B	4
4.2 Image Generation: SDXL + Refiner	4
4.3 Prompt Engineering	5
4.4 Optional Video: LTX Video	5
5. AMD Radeon and ROCm Adaptation	5
5.1 Hardware Platform	5
5.2 ROCm-Specific Optimizations	5
5.3 Compatibility Considerations	6
6. Uniqueness Engine	6
7. 4K Comic Page Assembly	7
8. Technical Summary and Innovation	7

1. Project Overview

Textovid is a multimodal AI application purpose-built for the AMD AI DevMaster Hackathon 2026, competing in Track 1: Development of Multimodal Content Creation Tools. The application generates complete, original comic books from either free-text story prompts or structured category selections, combining large language model storytelling with state-of-the-art image generation and professional comic page layout assembly. Every stage of the pipeline is designed to showcase the capabilities of AMD Radeon GPUs running ROCm, from the high-throughput SDXL inference engine to the memory-optimized refiner pass that produces 4K-quality panel artwork.

The system operates through a four-stage pipeline: story generation via the free Qwen 35B LLM API (consuming zero GPU credits), followed by SDXL-based panel image generation on the AMD Radeon GPU, optional LTX Video panel animation, and finally a CPU-based comic layout engine that assembles all panels into polished 4K comic pages complete with speech bubbles, narration boxes, sound-effect text, and a professional title page. The entire workflow is orchestrated through a cyberpunk-themed Gradio web interface featuring a 4K video background, animated grid overlays, and neon HUD-style elements.

A key differentiator is the Uniqueness Engine, a combinatorial randomization system that draws from a library of 20 plot structures, 19 character archetypes, 30 unique character traits, 28 surreal settings, and 20 narrative twists. Each generated comic receives a SHA-256 content fingerprint proving its structural distinctness. This ensures that every submission and demo run produces genuinely original creative output, rather than variations on a template.

2. Target Users and Use Cases

2.1 Primary Audience

Textovid is designed for creative professionals, hobbyists, and educators who want to produce high-quality comic content without traditional artistic skills. The primary audience includes independent comic creators seeking rapid prototyping of story ideas, content marketers looking to create engaging visual narratives for social media, educators developing visual teaching materials, and game designers exploring narrative concepts through sequential art. The Gradio web interface makes the tool accessible to non-technical users, while the underlying Python modules provide a clean API for programmatic integration into larger creative workflows.

2.2 Creative Scenarios

The application supports a wide range of creative scenarios spanning ten genres (Superhero, Horror, Romance, Sci-Fi, Fantasy, Mystery, Comedy, Drama, Thriller, and Slice-of-Life) and twelve sub-genres including Cyberpunk, Steampunk, Solarpunk, and Post-Apocalyptic. Users can choose from eleven distinct art styles, ranging from Manga and Western Comic to Watercolor Illustration, Pixel Art, Film Noir, Hyperrealistic Digital Art, and Art Nouveau. Each art style is mapped to a carefully tuned set of image-generation prompt keywords that guide SDXL toward the desired aesthetic, ensuring visual consistency across all panels in a

single comic. The mood system (Dark, Epic, Intimate, Surreal, Gritty, and more) further influences lighting, composition, and color palette in the generated artwork.

2.3 Practical Value

Beyond creative entertainment, Textovid demonstrates significant practical value in several domains. In education, teachers can generate visual stories that make abstract concepts tangible for students. In marketing, brands can rapidly produce comic-style advertisements and social media content. In game development, narrative designers can visualize dialogue sequences and character interactions before investing in manual illustration. The system also serves as a compelling demonstration of AMD GPU capabilities for multimodal AI workloads, potentially driving adoption among AI developers who currently rely exclusively on NVIDIA hardware. The built-in GPU benchmark module provides quantifiable performance metrics that can be used for direct hardware comparisons.

3. System Architecture

3.1 Pipeline Overview

The Textovid pipeline follows a sequential four-stage architecture designed to maximize both quality and hardware utilization. The stages are: Story Generation, Image Generation, Optional Video Animation, and Comic Page Assembly. Each stage is implemented as an independent module with well-defined interfaces, making the system modular and maintainable. The story engine communicates with the free Qwen 35B API on Radeon Cloud, consuming zero GPU resources. This architectural decision means the entire GPU budget is reserved for the computationally intensive image generation stage, maximizing throughput.

Table 1: Pipeline Stages and Resource Allocation

Stage	Module	Compute	Output
1. Story Generation	story_engine.py	Cloud API (zero GPU)	Structured JSON script
2. Image Generation	image_engine.py	AMD Radeon GPU (ROCm)	1024x1024 or 2048x2048 panels
3. Video Animation	video_engine.py	AMD Radeon GPU (optional)	MP4 video clips
4. Page Assembly	comic_layout.py	CPU (Pillow/PIL)	2480x3508 PNG pages

3.2 Module Breakdown

The codebase consists of thirteen files totaling approximately 2,887 lines of Python, CSS, and shell script code. Each module handles a specific concern within the pipeline. The **config.py** module centralizes all configuration constants including model IDs, generation parameters, comic page dimensions (2480x3508 pixels for 4K output), and user-facing dropdown options for genres, art styles, moods, and themes. The

uniqueness.py module implements the combinatorial randomization engine with its extensive libraries of plot structures, character archetypes, traits, settings, and narrative twists. The **story_engine.py** module handles LLM communication, including a carefully crafted system prompt that instructs the model to produce self-contained visual descriptions for each panel, along with robust JSON extraction that handles common LLM output formatting issues.

Table 2: Project File Structure

File	Lines	Responsibility
app.py	~496	Main Gradio UI and pipeline orchestrator
config.py	~87	Centralized configuration and constants
story_engine.py	~228	LLM story generation via Qwen 35B API
image_engine.py	~254	SDXL + Refiner image generation on ROCm
video_engine.py	~134	LTX Video panel animation (optional)
comic_layout.py	~527	4K comic page composition and assembly
uniqueness.py	~251	Combinatorial randomization engine
gpu_utils.py	~124	GPU detection, VRAM monitoring, benchmarking
style.css	~200	Cyberpunk sci-fi UI theme

4. Model and Algorithm Introduction

4.1 Text Generation: Qwen 35B

The story generation stage leverages the Qwen3.6-35B-A3B model, a 35-billion-parameter large language model accessed through the free Radeon Cloud Model API. This choice is strategically significant for two reasons: first, it consumes zero GPU credits on the Radeon Cloud platform, reserving the entire GPU budget for image generation; second, the model produces highly creative and structurally coherent comic scripts when guided by the carefully engineered system prompt. The system prompt instructs the model to generate self-contained visual descriptions for each panel, specifying composition, camera angle, lighting, colors, character poses, and environment details in 60-100 words per panel. The JSON output schema includes hierarchical structure with pages, panels, dialogue, narration, and sound effects. A temperature of 0.92 balances creativity with coherence, producing stories that are surprising yet narratively sound.

4.2 Image Generation: SDXL + Refiner

Panel artwork is generated using Stability AI's Stable Diffusion XL Base 1.0 model, with an optional High-Res Fix pass through the SDXL Refiner 1.0 model. The image generation engine implements a two-pass approach:

the first pass generates a 1024x1024 pixel base image using the DPM-Solver++ multistep scheduler with Karras sigmas, which produces high-quality results in fewer inference steps compared to the default Euler scheduler. When High-Res Fix is enabled (the default), the base image is upsampled to 2048x2048 pixels using LANCZOS resampling, then passed through the SDXL Refiner with a reduced denoising strength of 0.35 to add fine detail without significantly altering the composition established in the base pass. This two-pass approach produces panels with sharp details suitable for 4K comic page assembly at 2480x3508 pixel output resolution.

4.3 Prompt Engineering

The image prompt construction system combines multiple elements into a single optimized prompt for each panel. The **build_image_prompt()** function assembles the scene description (from the LLM-generated script), art-style-specific keywords from the Uniqueness Engine's style mapping (11 art styles with curated prompt modifiers), mood-specific lighting and atmosphere keywords (10 mood mappings), character visual descriptions (up to 2 characters to maintain prompt focus), and universal quality boosters such as "masterpiece, best quality, highly detailed, sharp focus, professional comic book panel, cinematic composition." The negative prompt filters common artifacts including blur, deformation, extra fingers, watermarks, and JPEG compression artifacts.

4.4 Optional Video: LTX Video

For users who want to add motion to their comic panels, Textovid includes an optional video animation stage powered by Lightricks' LTX-Video-0.9.1-dev model. This feature is disabled by default and can be enabled through an environment variable or a checkbox in the advanced settings panel. When activated, up to three panels are animated into short video clips (25 frames at 8 FPS, approximately 3 seconds each) using image-to-video generation. The video pipeline supports CPU offloading for memory management and saves output as MP4 files via the imageio library. This feature demonstrates the versatility of the AMD Radeon GPU for different modalities of AI content generation beyond static images.

5. AMD Radeon and ROCm Adaptation

5.1 Hardware Platform

Textovid is developed and optimized for the AMD Radeon RX 7900 XTX GPU with 24 GB of GDDR6 VRAM, running on the Radeon Cloud platform. The software stack is built on PyTorch 2.11.0 with the ROCm 7.2 backend, which provides the HIP (Heterogeneous-compute Interface for Portability) layer that translates CUDA-like operations to AMD hardware. The application is deployed via Gradio 6.20.0, which serves the web interface on the Radeon Cloud JupyterLab instance. All dependencies (diffusers, transformers, PIL, requests) are pre-installed in the /opt/venv/ virtual environment provided by the platform, eliminating the need for manual dependency installation.

5.2 ROCm-Specific Optimizations

Table 3: AMD ROCm Optimization Details

Optimization	Implementation	Impact
Float16 Inference	<code>torch_dtype=torch.float16</code> for all pipeline operations	2x speedup, 50% VRAM reduction
Karras Noise Schedule	<code>use_karras_sigmas=True</code> in DPM-Solver++ scheduler	Better quality in fewer steps
Memory-Efficient Attention	<code>enable_xformers_memory_efficient_attention()</code> with fallback	Reduced VRAM peak usage
SDXL Refiner Hi-Res Fix	Two-pass: base 1024px then refiner 2048px at 0.35 denoise	4K-quality panel artwork
Zero-Cost LLM	Qwen 35B via cloud API, no local GPU inference	Entire GPU budget for images
VRAM Monitoring	<code>torch.cuda.mem_get_info()</code> + <code>get_device_properties()</code>	Real-time memory tracking
Dynamic Layout	Non-uniform grid randomization per page (3-6 panels)	Visual variety without user input

5.3 Compatibility Considerations

During development, several ROCm-specific compatibility issues were identified and resolved. The PyTorch 2.11.0 API renamed the device property from **total_mem** to **total_memory**, requiring a compatibility shim using `getattr()` with a fallback to `mem_get_info()` for total VRAM calculation. Gradio 6.20.0 moved the theme and CSS parameters from the `Blocks()` constructor to the `launch()` method, requiring an update to the application launch code. The xformers memory-efficient attention module may not be available in all ROCm builds, so the implementation includes a try/except fallback to default attention. The install script attempts multiple ROCm wheel versions (6.2, 6.1, 6.0) to maximize compatibility across different Radeon Cloud instance configurations.

6. Uniqueness Engine

The Uniqueness Engine is a combinatorial creative system designed to ensure that every comic generated by Textovid is structurally and narratively distinct. Unlike simple random seed variation (which produces superficially different images from the same prompt), the Uniqueness Engine randomizes the fundamental creative building blocks of the story itself. It draws from five independent libraries: 20 plot structures (including Hero's Journey, Rashomon, Circular, Frame Narrative, and Bait-and-Switch), 19 character archetypes (Reluctant Hero, AI Awakening, Shape-Shifter with Identity Crisis, and more), 30 unique character traits (each a vivid, story-generating detail such as "bleeds ink instead of blood" or "can hear the emotions of inanimate objects"), 28 surreal settings (a city built inside the ribcage of a colossal fossil, a frozen ocean with

cities beneath the ice), and 20 narrative twists.

The engine operates by randomly selecting one element from each library and combining them into a unified creative premise. Settings are mashed up in pairs (two randomly selected settings are combined) to create truly novel world-building concepts. Each premise is hashed using SHA-256 to produce a 16-character fingerprint that serves as a cryptographic proof of uniqueness. This fingerprint is displayed in the generation status report, allowing judges and users to verify that different runs produce fundamentally different creative foundations. The premise is then passed as a directive to the LLM, which uses it as a creative constraint when generating the full comic script. This approach ensures that even if two users select the same genre and mood, the resulting comics will differ dramatically in plot, characters, and setting.

7. 4K Comic Page Assembly

The comic layout engine is responsible for transforming the raw panel images and script data into polished, publication-ready comic pages. The engine operates at 2480x3508 pixel resolution (A4 at 210 DPI), which meets the 4K quality threshold for print-ready output. Each page is composed through a multi-step process that includes page border rendering, header bar with comic title and page numbering, dynamic grid layout calculation, panel image placement with LANCZOS resampling, narration box rendering (yellow caption boxes at the top of panels), speech bubble drawing (rounded rectangles with directional tails and speaker labels), and stylized SFX text rendering with outline effects for impact words like "CRASH" or "WHOOOOSH."

The title page deserves special mention as a showcase feature. It uses the first generated panel image as a full-bleed background, applies a brightness reduction overlay (35% brightness) for text readability, and adds a vertical gradient overlay from transparent to near-opaque dark blue at the bottom. The comic title is rendered in large bold text with a cyan glow effect (achieved through multiple offset shadow passes), the synopsis is displayed as a subtitle, and a branded footer reads "GENERATED BY TEXTTOVID" in spaced cyan text. The dynamic grid layout system generates non-uniform panel arrangements (wide-top, feature-bottom, cinematic split, and more) for each page, ensuring visual variety across the entire comic without user intervention.

8. Technical Summary and Innovation

Textovid represents a comprehensive demonstration of multimodal AI content creation on AMD Radeon hardware. The system integrates four distinct AI capabilities (text generation, image generation, video generation, and intelligent layout) into a seamless end-to-end pipeline. The technical innovations include the zero-cost LLM architecture (offloading story generation to a free cloud API), the two-pass High-Res Fix pipeline that produces 4K-quality panels from a consumer GPU, the combinatorial Uniqueness Engine with cryptographic fingerprinting, and the dynamic comic layout system that produces professional-grade visual output without manual design intervention. All of this is wrapped in a visually striking cyberpunk-themed web interface that makes the technology accessible and engaging for both technical and non-technical users.

The AMD-specific optimizations demonstrate a deep understanding of the ROCm ecosystem: float16 inference for throughput, Karras sigmas for quality-per-step efficiency, xformers for memory reduction, and careful VRAM management that keeps the full SDXL + Refiner pipeline within the 24 GB budget of the Radeon RX 7900 XTX. The built-in benchmark module provides quantifiable performance data that can be used for direct comparisons, making this submission not only a creative tool but also a performance showcase for AMD GPU capabilities in real-world multimodal AI workloads. The project is fully open-source, well-documented, and designed for reproducibility on any Radeon Cloud instance with a compatible GPU.